

ECOM90009 Quantitative Methods for Business

Workshop 9 (Week 10)

Zheng Fan

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Introduction

Zheng Fan

- Know more about myself: zhengfan.site
- Ph.D student in Economics at Unimelb

If you need any help, don't be shy

- Weekly consultation & Ed discussion board (your **first priority**)
- Email: jnlye@unimelb.edu.au for subject related issues
- Consult Stop 1 for special consideration
- Email: fan.z@unimelb.edu.au

Before asking any questions, make sure you have gone through the materials on Canvas and existing posts on Ed discussion board!

Assignment 3: Reminder

Assignment 3 is now available on Canvas. This assignment contributes up to 10% of your final grade for this subject.

Group must be registered by: Thursday, May 21, at 11:59 pm

Submission Deadline: Thursday, May 28, at 11:59 pm.

Introduction

- Regression diagnostics
 - Calculating and interpreting R^2
- Constructing point predictions
- Using Excel to generate regression output
- Introduction to Multiple regression model

Question 1

Indicate whether each statement below is true or false. Justify your answer.

- a When the actual values y of a dependent variable and the corresponding predicted values $\hat{y}$ are the same, the standard error of estimate, s_e , will be -1.0 .

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- What is s_e ?

Question 1

SSR: sum of squared regression
SST: sum of squared total
SSE: sum of squared residual

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- What is s_ϵ ?

In a simple regression model,

$$s_\epsilon = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{\sum e_i^2}{n-2}}$$

where $\sum e_i^2 = \sum (y_i - \hat{y}_i)^2$

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- When $y_i = \hat{y}_i$, what must $\sum e_i^2$ be?

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- If the error variable ε is normally distributed, the test statistic for testing $H_0 : \beta_1 = 0$ in a simple linear regression follows the Student t -distribution with $n - 1$ degrees of freedom.
- If the error variable ε is normally distributed, the OLS estimator $\hat{\beta}_j$ is a normal random variable.
- Since the OLS estimator is normally distributed, $z = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\text{var}(\hat{\beta}_j)}} \sim N(0, 1)$
- However, we do not know σ^2 and hence do not know $\text{var}(\hat{\beta}_j)$.
- We use $t = \frac{\hat{\beta}_j - \beta_j}{\text{se}(\hat{\beta}_j)} \sim t_{n-2}$

FALSE statement

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- Given that $SSE = 150$ and $SSR = 450$, the proportion of the variation in y that is explained by the variation in x is 0.75.

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- What is the name of the statistical measurement that examines the proportion of variation in y that is explained by the variation in x ?

Question 1

- Given that $SSE = 150$ and $SSR = 450$, the proportion of the variation in y that is explained by the variation in x is 0.75.
- What is the name of the statistical measurement that examines the proportion of variation in y that is explained by the variation in x ?
- $R^2 = 1 - \frac{SSE}{SST}$ where $SSE = 150$ and $SST = SSR + SSE = 450 + 150 = 600$.

$$\therefore R^2 = 1 - \frac{150}{600} = 1 - 0.25 = 0.75$$

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- $R^2 = 1 - \frac{SSE}{SST}$ where $SSE = 150$ and $SST = SSR + SSE = 450 + 150 = 600$.

$$\therefore R^2 = 1 - \frac{150}{600} = 1 - 0.25 = 0.75$$

- Alternatively, $R^2 = \frac{SSR}{SST} = \frac{450}{600} = 0.75$

TRUE statement

Question 1

- d An outlier is an observation that is unusually small or unusually large.

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- TRUE. The definition of an outlier is exactly what the statement is, i.e. an outlier is an observation that is unusually small or unusually large.

Question 2

The following table contains the data taken on number of weekly study hours (HOURS) and the final marks (MARKS) obtained by students in a quantitative methods for business subject.

<i>Student</i>	MARKS	HOURS
1	58	21
2	69	24
3	62	26
4	73	27
5	74	29
6	62	25
7	55	25
8	76	30

- a Estimate the relationship between MARKS and HOURS using OLS; that is, obtain the **intercept** and **slope estimates** in the regression of MARKS on HOURS. Perform all calculations by hand. Does the intercept have a useful interpretation here? Explain. How much higher is MARKS predicted to be if the HOURS is increased by five units? [Use the workshop09-1.xls file.](#)

Let $y_i = \text{MARKS}_i$ and $x_i = \text{HOURS}_i$

Recall:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Let $y_i = \text{MARKS}_i$ and $x_i = \text{HOURS}_i$

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	y_i	x_i	$(y_i - \bar{y})$	$(x_i - \bar{x})$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
	58	21	-8.125	-4.875	66.01563	23.76563	39.60938
	69	24	2.875	-1.875	8.265625	3.515625	-5.39063
	62	26	-4.125	0.125	17.01563	0.015625	-0.51563
	73	27	6.875	1.125	47.26563	1.265625	7.734375
	74	29	7.875	3.125	62.01563	9.765625	24.60938
	62	25	-4.125	-0.875	17.01563	0.765625	3.609375
	55	25	-11.125	-0.875	123.7656	0.765625	9.734375
	76	30	9.875	4.125	97.51563	17.01563	40.73438
Sum	529	207	0	0	438.875	56.875	120.125

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Sum	529	207	0	0	438.875	56.875	120.125

$$\bar{x} = \frac{\sum_{i=1}^8 x_i}{n} = \frac{207}{8} = 25.875, \quad \bar{y} = \frac{\sum_{i=1}^8 y_i}{n} = \frac{529}{8} = 66.125$$

$$\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = 120.125, \quad \sum_{i=1}^n (x_i - \bar{x})^2 = 56.875, \quad \bar{x} = 25.875, \quad \bar{y} = 66.125$$

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$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 66.125 - (2.112 \times 25.875) = 11.447$$

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- Does the intercept have a useful interpretation? Explain.

The intercept does not have a useful interpretation because the value of the variable HOURS is not close to zero for the population of interest.

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- How much higher is MARKS predicted to be if the HOURS is increased by five units?

If HOURS is 5 units higher, on average, MARKS increases by $2.112 \times 5 = 10.56$

Alternative way of computing slope coefficient

y_i	x_i	$x_i y_i$	x_i^2
58	21	1218	441
69	24	1656	576
62	26	1612	676
73	27	1971	729
74	29	2146	841
62	25	1550	625
55	25	1375	625
76	30	2280	900
Sum	529	207	13808
			5413

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2} && \text{From last week Q5} \\ &= \frac{13808 - 8(25.875)(66.125)}{5413 - 8(25.875)^2} = \frac{120.125}{56.875} = 2.112\end{aligned}$$

- Compute the fitted values, residuals and the squared residuals for each observation.

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Previously, our fitted regression line is $\hat{y} = 11.447 + 2.112 x$

Residuals: $e_i = y_i - \hat{y}$

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Residuals: $e_i = y_i - \hat{y}_i$

y_i	x_i	$\hat{y}_i = 11.447 + 2.112 x$	$e_i = y_i - \hat{y}_i$	e_i^2
58	21	55.82857	2.171429	4.715102
69	24	62.16484	6.835165	46.71948
62	26	66.38901	-4.38901	19.26342
73	27	68.5011	4.498901	20.24011
74	29	72.72527	1.274725	1.624925
62	25	64.27692	-2.27692	5.184379
55	25	64.27692	-9.27692	86.0613
76	30	74.83736	1.162637	1.351726
Sum			-7.1E-15	185.1604

- What is the predicted value of MARKS when HOURS = 25?

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When $x = 25$, $\hat{y} = 11.447 + 2.112(25) = 64.25$

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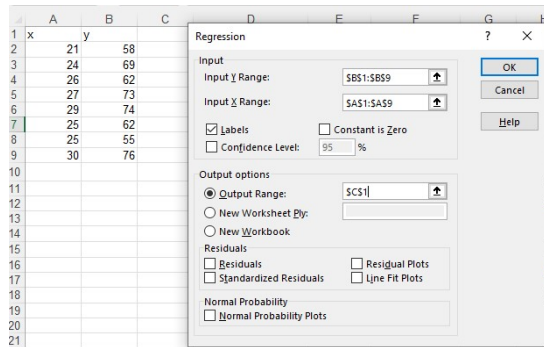
Recall that $SSE = \sum_{i=1}^n e_i^2$ and $SST = \sum_{i=1}^n (y_i - \bar{y})^2$

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} = 1 - \frac{185.1604}{438.875} = 0.5781$$

About 57.8% of the variation in MARKS is explained by HOURS in this small sample of students.

- e Check your answers in Excel using the “regression” option in the Data analysis tool. The data is in the sheet labelled Q1 in the Excel file Workshop09.xlsx.

Using Excel data analysis tool



A	B	C	D	E	F	G	H	I
x	y	SUMMARY OUTPUT						
21	58							
24	69	Regression Statistics						
26	62	Multiple R	0.7603303					
27	73	R Square	0.5781021					
29	74	Adjusted R Square	0.5077858					
25	62	Standard Error	5.5551844					
25	55	Observations	8					
30	76							
		ANOVA						
			df	SS	MS	F	Significance F	
		Regression	1	253.7145604	253.7146	8.22145	0.02852744	
		Residual	6	185.1604396	30.86007			
		Total	7	438.875				

Question 3

Several years ago, Coca-Cola attempted to change its 100-year-old recipe. One reason why company's management thought this was necessary was competition from Pepsi cola.

Surveys of Pepsi drinkers indicated that they preferred Pepsi because it was sweeter than Coke. The management of Coca Cola performed extensive surveys in which consumers tasted various versions of the new coke.

Suppose that a random sample of 200 Cola drinkers were given versions of Coke with different amounts of sugar. After tasting the product, each drinker was asked to rate the taste quality. The possible responses were as follows: 1 = poor, 2 = fair, 3 = average, 4 = good and 5 = excellent. The responses (rating and denoted y) and the sugar content (percentage by volume and denoted x) were both recorded.

Given the sample statistics from the survey produced in the Table below, can management infer at the 5% level of significance that sugar content affects drinkers' ratings of the cola?

$n = 200$	$\bar{x} = 12.305$	$\bar{y} = 2.91$
$s_x^2 = 37.00701$	$s_{xy} = 6.56528$	$SSE = 138.6005$

Sample statistics

$n = 200$	$\bar{x} = 12.305$	$\bar{y} = 2.91$
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$$\hat{\beta}_1 = \frac{s_{xy}}{s_x^2} = \frac{6.56528}{37.00701} = 0.17741$$

$$\text{rating} = \beta_0 + \beta_1 \text{ sugar content} + \varepsilon$$

Step 1: Is there any relationship between sugar content and ratings?

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Step 1: Is there any relationship between sugar content and ratings?

$$H_0 : \beta_1 = 0 \qquad H_A : \beta_1 \neq 0$$

Sample statistics

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Step 1: Is there any relationship between sugar content and ratings?

$$H_0 : \beta_1 = 0 \qquad H_A : \beta_1 \neq 0$$

Step 2: Assuming that the required conditions for population ε_i hold (1. Probability distribution of ε_i is Normal; 2. $E(\varepsilon) = 0$; 3. Homoscedasticity; 4. no autocorrelation and 5. no endogeneity), then we can use:

$$t = \frac{\hat{\beta}_1 - \beta_1}{s_{\hat{\beta}_1}} \sim t_{n-2}$$

Step 3: Use $\alpha = 0.05$

Step 4: Reject H_0 if $t < -1.972$ or $t > 1.972$ ($df = 198$)

Excel function: `=T.INV.2T(0.05,198)` which gives 1.972017478

Or look up the critical values of the Student t Distribution table

Critical Values of the Student t Distribution - upper (right) tail only

Degrees of Freedom	t _{0.10}	t _{0.05}	t _{0.025}	t _{0.01}	t _{0.005}
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
35	1.306	1.690	2.030	2.438	2.724
40	1.303	1.684	2.021	2.423	2.704
45	1.301	1.679	2.014	2.412	2.690
50	1.299	1.676	2.009	2.403	2.678
60	1.296	1.671	2.000	2.390	2.660
70	1.294	1.667	1.994	2.381	2.648

Sample statistics

$n = 200$	$\bar{x} = 12.305$	$\bar{y} = 2.91$
$s_x^2 = 37.00701$	$s_{xy} = 6.56528$	$SSE = 138.6005$

Step 5: $t = \frac{\hat{\beta}_1 - \beta_1}{s_{\hat{\beta}_1}}$

Need to calculate $s_{\hat{\beta}_1} = \frac{s_\varepsilon}{\sqrt{(n-1)s_x^2}}$ where

$$s_\varepsilon = \sqrt{\frac{SSE}{200 - 2}} = \sqrt{\frac{138.6005}{198}} = 0.836662$$

$$t = \frac{\hat{\beta}_1 - \beta_1}{s_{\hat{\beta}_1}} = \frac{0.177406 - 0}{0.836662 / \sqrt{199 \times 37.00701}} = 18.197$$

Step 6: Since $18.197 > 1.972$, we reject H_0 at the 5% level of significance.

There is sufficient evidence at the 5% level to support the alternative hypothesis that sugar content affects drinkers' ratings of the Cola.

Question 4

The Excel spreadsheet Workshop09.xlsx contains data on used cars. All cars in this data set were less than one year old when priced and considered to be in excellent condition. The variable price is the suggested retail price of the used car and mileage is the number of miles the car has been driven.

For this question, do all the estimation using the regression option in the Data analysis tool in Excel.

- a Estimate the simple regression model, $\text{price} = \beta_0 + \beta_1 \text{mileage} + \varepsilon$ and interpret the estimated coefficient on mileage. How does price change if two cars are identical except one has 50,000 more miles?
- b Divide the sample by the variable cylinder which is the number of cylinders in the engine - 4, 6 or 8. Re-estimate the simple regression model in (a) above for each of the subsamples. For each subsample, interpret the estimated coefficient on mileage. For each subsample, how does price change if two cars are identical except one has 50,000 more miles?

- a Estimate the simple regression model, $\text{price} = \beta_0 + \beta_1 \text{mileage} + \varepsilon$ and interpret the estimated coefficient on mileage. How does price change if two cars are identical except one has 50,000 more miles?

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.1430505					
R Square	0.0204634					
Adjusted R Square	0.0192421					
Standard Error	9789.2883					
Observations	804					
ANOVA						
	df	SS	MS	F	Significance F	
Regression	1	1605590375	1.606E+09	16.75454	4.6847E-05	
Residual	802	76855792486	95830165			
Total	803	78461382861				
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	24764.559	904.3632777	27.383419	4.47E-117	22989.36052	26539.757
Mileage	-0.172521	0.042147756	-4.093231	4.685E-05	-0.255253442	-0.089788

- a Estimate the simple regression model, $\text{price} = \beta_0 + \beta_1 \text{mileage} + \varepsilon$ and interpret the estimated coefficient on mileage. How does price change if two cars are identical except one has 50,000 more miles?

SUMMARY OUTPUT						
<i>Regression Statistics</i>						
Multiple R	0.1430505					
R Square	0.0204634					
Adjusted R Square	0.0192421					
Standard Error	9789.2883					
Observations	804					
<i>ANOVA</i>						
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>	
Regression	1	1605590375	1.606E+09	16.75454	4.6847E-05	
Residual	802	76855792486	95830165			
Total	803	78461382861				
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	24764.559	904.3632777	27.383419	4.47E-117	22989.36052	26539.757
Mileage	-0.172521	0.042147756	-4.093231	4.685E-05	-0.255253442	-0.089788

- Slope coefficient: For every extra mile driven, the suggested retail price of a used car decreases on average by \$0.17 (17 cents).
- Difference in price: If two cars are identical except one has 50,000 more miles, then it would be valued at around $-0.173 \times 50,000 \approx \$8,650$ less.

- b Divide the sample by the variable `cylinder` which is the number of cylinders in the engine - 4, 6 or 8. Re-estimate the simple regression model in (a) above for each of the subsamples. For each subsample, interpret the estimated coefficient on `mileage`. For each subsample, how does price change if two cars are identical except one has 50,000 more miles?

4 Cylinder		Cylinder	6 Cylinder		Cylinder	8 Cylinder		Cylinder
Price	Mileage		Price	Mileage		Price	Mileage	
12146.19	10011	4	17314.1	8221	6	51154.05	2202	8
12163.82	12101	4	17542.04	9135	6	49248.16	6685	8
11472.02	19699	4	16218.85	13196	6	46747.67	15343	8
11017.17	20100	4	16336.91	16342	6	44130.62	21341	8
11096.86	20334	4	16339.17	19832	6	44084.91	21367	8
10386.04	22225	4	15709.05	22236	6	43892.47	23371	8
11137.05	22484	4	15230	22576	6	44300.64	23751	8
11045.11	24568	4	15048.04	22964	6	42677.6	24052	8
10777.05	27906	4	14862.09	24021	6	43374.05	25199	8
9928.188	29680	4	15295.02	27325	6	40619.07	30082	8
12649.11	3629	4	21335.85	10237	6	39801.55	14095	8
12314.59	4142	4	20538.09	15066	6	40335.74	14743	8
11318.01	11156	4	20512.09	16633	6	39307.01	16041	8
12409.95	11981	4	19924.16	19800	6	38600.24	17138	8
11555.27	13404	4	19774.25	23359	6	38445.9	18661	8
11700.11	15253	4	19344.17	23765	6	36077.8	21966	8
11215.02	19945	4	19105.13	24008	6	35866.58	24415	8
10144.95	23963	4	18543.43	26034	6	35338.65	25163	8
10491.08	30948	4	17808.2	32896	6	36154.3	25339	8
9954.054	37345	4	17968.84	34665	6	34685.66	25421	8
11031.13	20156	4	22358.88	8970	6	42820.33	5499	8
11343.05	20186	4	23785.92	10577	6	41378.05	8125	8
11391.21	21421	4	22926.09	14363	6	40856.39	12791	8

4 cylinders

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.082544					
R Square	0.006813					
Adjusted R Squ	0.00428					
Standard Error	7814.208					
Observations	394					
ANOVA						
	df	SS	MS	F	Significance F	
Regression	1	164207376.3	1.64E+08	2.689198	0.101832869	
Residual	392	23936245209	61061850			
Total	393	24100452586				
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	19438.85	1038.714923	18.71433	2.67E-56	17396.70344	21481
Mileage	-0.07839	0.047802451	-1.63988	0.101833	-0.172371404	0.015591

- Slope coefficient: For every extra mile driven, the suggested retail price of a used car (with 4 cylinders) decreases on average by \$0.08 (8 cents).
- Difference in price: If two cars are identical except one has 50,000 more miles, then it would be valued at around $-0.078 \times 50,000 \approx \$3,900$ less.

6 cylinders

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.276222454					
R Square	0.076298844					
Adjusted R Square	0.073299814					
Standard Error	4458.25928					
Observations	310					
ANOVA						
	df	SS	MS	F	Significance F	
Regression	1	505670748.8	5.06E+08	25.44118	7.80435E-07	
Residual	308	6121831348	19876076			
Total	309	6627502097				
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	23251.08119	677.5128278	34.31829	2.9E-107	21917.9419	24584.22047
Mileage	-0.16202001	0.032121815	-5.04392	7.8E-07	-0.22522598	-0.098814048

- Slope coefficient: For every extra mile driven, the suggested retail price of a used car (with 6 cylinders) decreases on average by \$0.16 (16 cents).
- Difference in price: If two cars are identical except one has 50,000 more miles, then it would be valued at around $-0.162 \times 50,000 \approx \$8,100$ less.

8 cylinders

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.354482392					
R Square	0.125657766					
Adjusted R Square	0.116735907					
Standard Error	10086.47456					
Observations	100					
ANOVA						
	df	SS	MS	F	Significance F	
Regression	1	1.43E+09	1.43E+09	14.08426	0.00029631	
Residual	98	9.97E+09	1.02E+08			
Total	99	1.14E+10				
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	47304.39685	2439.588	19.39033	2.58E-35	42463.1148	52145.67894
Mileage	-0.425860186	0.113475	-3.7529	0.000296	-0.65104756	-0.200672808

- Slope coefficient: For every extra mile driven, the suggested retail price of a used car (with 8 cylinders) decreases on average by \$0.43 (43 cents).
- Difference in price: If two cars are identical except one has 50,000 more miles, then it would be valued at around $-0.426 \times 50,000 \approx \$21,300$ less.

- g What do you conclude from comparing the simple regression results on the full sample to the simple regression results on subsamples based on the number of cylinders?

$$\text{price} = \beta_0 + \beta_1 \text{ mileage} + \varepsilon$$

Full sample	4 cylinders	6 cylinders	8 cylinders
\$8,650	\$3,900	\$8,100	\$21,300

- What do you conclude from comparing the simple regression results on the full sample to the simple regression results on subsamples based on the number of cylinders?

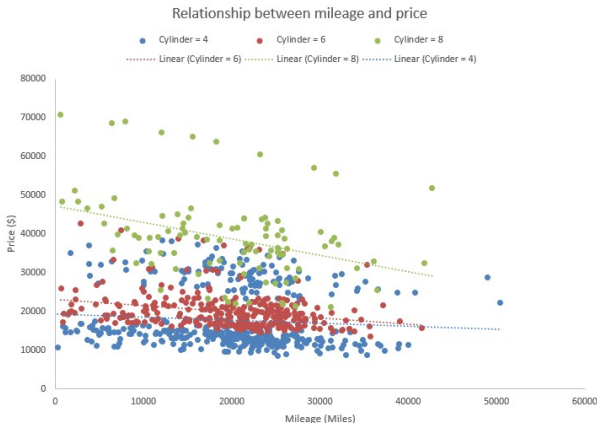
$$\text{price} = \beta_0 + \beta_1 \text{ mileage} + \varepsilon$$

Full sample	4 cylinders	6 cylinders	8 cylinders
\$8,650	\$3,900	\$8,100	\$21,300

- The number of cylinders of the engine of the car seems to make a difference.
- This suggests that there are other important explanatory variables that have been excluded from the regression model of the full sample.

Scatter plot of price and mileage for each cylinder size

The scatter plot below of price and mileage for each cylinder size highlights the different regression results from each subsample and also indicates that other important explanatory variables may be excluded from the simple regression model.



Question 5

A random sample of 100 observations was taken to estimate the regression model,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

A software package was employed to produce the output in the accompanying table. However, because of a printer malfunction some values are missing. Fill in the missing values that are labelled (a) to (e).

Source	d.f.	SS	MS
Regression		573.6	
Residual		352.3	
Total		925.9	

Question 5

A random sample of 100 observations was taken to estimate the regression model,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

A software package was employed to produce the output in the accompanying table. However, because of a printer malfunction some values are missing. Fill in the missing values that are labelled (a) to (e).

Source	d.f.	SS	MS
Regression	2	573.6	
Residual		352.3	
Total		925.9	

Question 5

A random sample of 100 observations was taken to estimate the regression model,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

A software package was employed to produce the output in the accompanying table. However, because of a printer malfunction some values are missing. Fill in the missing values that are labelled (a) to (e).

Source	d.f.	SS	MS
Regression	2	573.6	
Residual	97	352.3	
Total		925.9	

Question 5

A random sample of 100 observations was taken to estimate the regression model,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

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Source	d.f.	SS	MS
Regression	2	573.6	
Residual	97	352.3	
Total	99	925.9	

Question 5

A random sample of 100 observations was taken to estimate the regression model,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

A software package was employed to produce the output in the accompanying table. However, because of a printer malfunction some values are missing. Fill in the missing values that are labelled (a) to (e).

Source	d.f.	SS	MS
Regression	2	573.6	286.8
Residual	97	352.3	
Total	99	925.9	

Question 5

A random sample of 100 observations was taken to estimate the regression model,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

A software package was employed to produce the output in the accompanying table. However, because of a printer malfunction some values are missing. Fill in the missing values that are labelled (a) to (e).

Source	d.f.	SS	MS
Regression	2	573.6	286.8
Residual	97	352.3	3.63
Total	99	925.9	

Thanks for your attention!

Let me know if there is any questions!